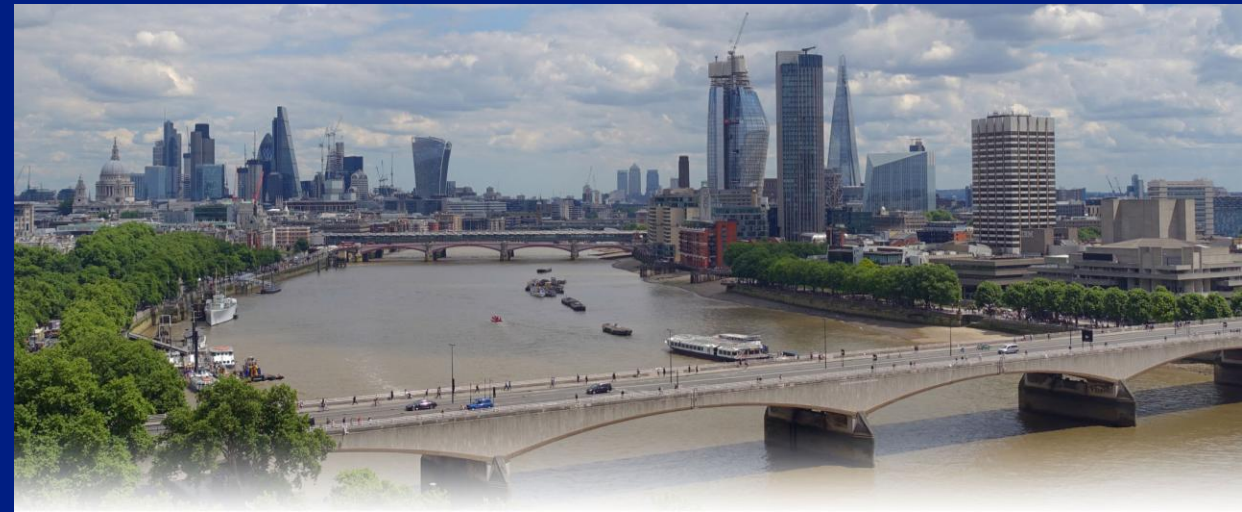


Lecture 12+13: The Macroeconomic Environment of Business



School for Business
and Society

Maryam Alhalboni PhD.
Maryam.Alhalboni @york.ac.uk



Lecture overview



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- The Macroeconomic Environment
- Unemployment
- Inflation
- Foreign Trade and Global Relationships
- Financial Well-being and Financialisation
- Economic Volatility and the Business Cycle
- The Circular Flow of Income
- Economic Growth
- Measuring National Income and Output

Macroeconomic variables

The environment is defined by five interrelated core variables:

- **Economic Growth:** The pace of expansion in national output.
- **Unemployment:** The utilization of the labor force.
- **Inflation:** The rate at which the general level of prices rises.
- **International Relationships:** Economic interactions with the rest of the world (trade/capital).
- **Financial Well-being:** The health of the financial system and its relationship with individuals, businesses, and government.

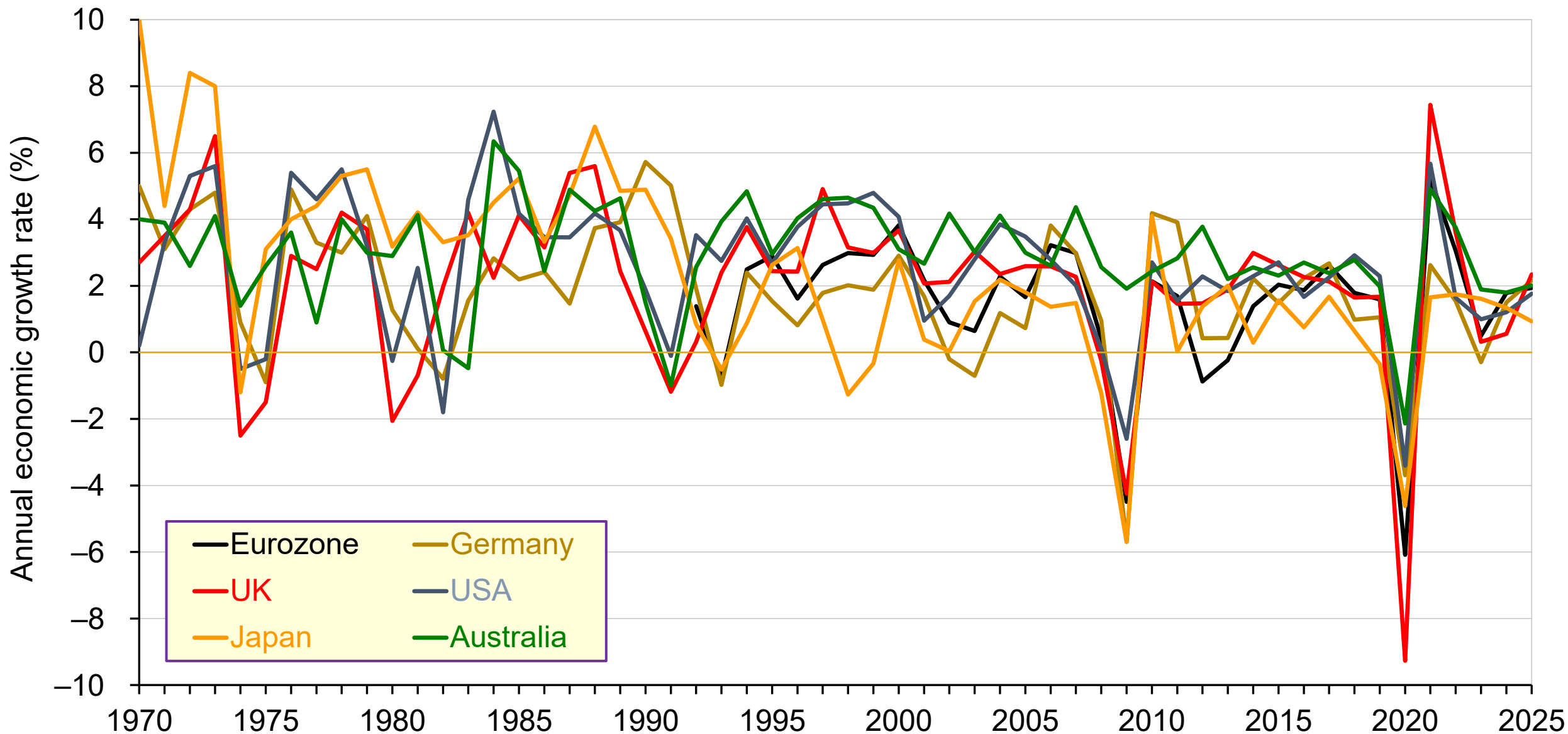
Measuring Growth (Nominal vs. Real GDP)

- Economic growth is the percentage change in an economy's output (GDP) over a specific period (typically quarterly or annually).
- Nominal GDP (Money GDP):
 - Measured using current prices (prices of the time).
 - Includes the effects of inflation; can be misleading if prices rise while volume stays flat.
- Real GDP:
 - Measured using constant prices from a specific base year.
 - Adjusts for inflation to show the true volume of output.
- Real GDP allows for accurate comparisons of actual production across different years.

Short term and the business cycle:

- Growth rates are rarely stable; economies experience constant fluctuations in GDP, unemployment, and inflation.
- The Business Cycle: The inherent instability of an economy, characterized by periods of:
 - Booms: High growth and activity.
 - Slowdowns/Recessions: Low or negative growth.
- The International Dimension:
 - Business cycles exist at national, regional (e.g., Eurozone), and global levels.
 - Growing Synchronicity: Increased globalization means national cycles are increasingly aligned with the International Business Cycle.

Annual economic growth rates



Notes: Data from 2022 based on forecasts; Eurozone = 19 countries using the euro (as of 2022); figures for Germany based on West Germany only up to 1991.
Sources: Based on data in *AMECO Database* (EC) up to 1980; and *World Economic Outlook* (IMF, October 2022); various forecasts

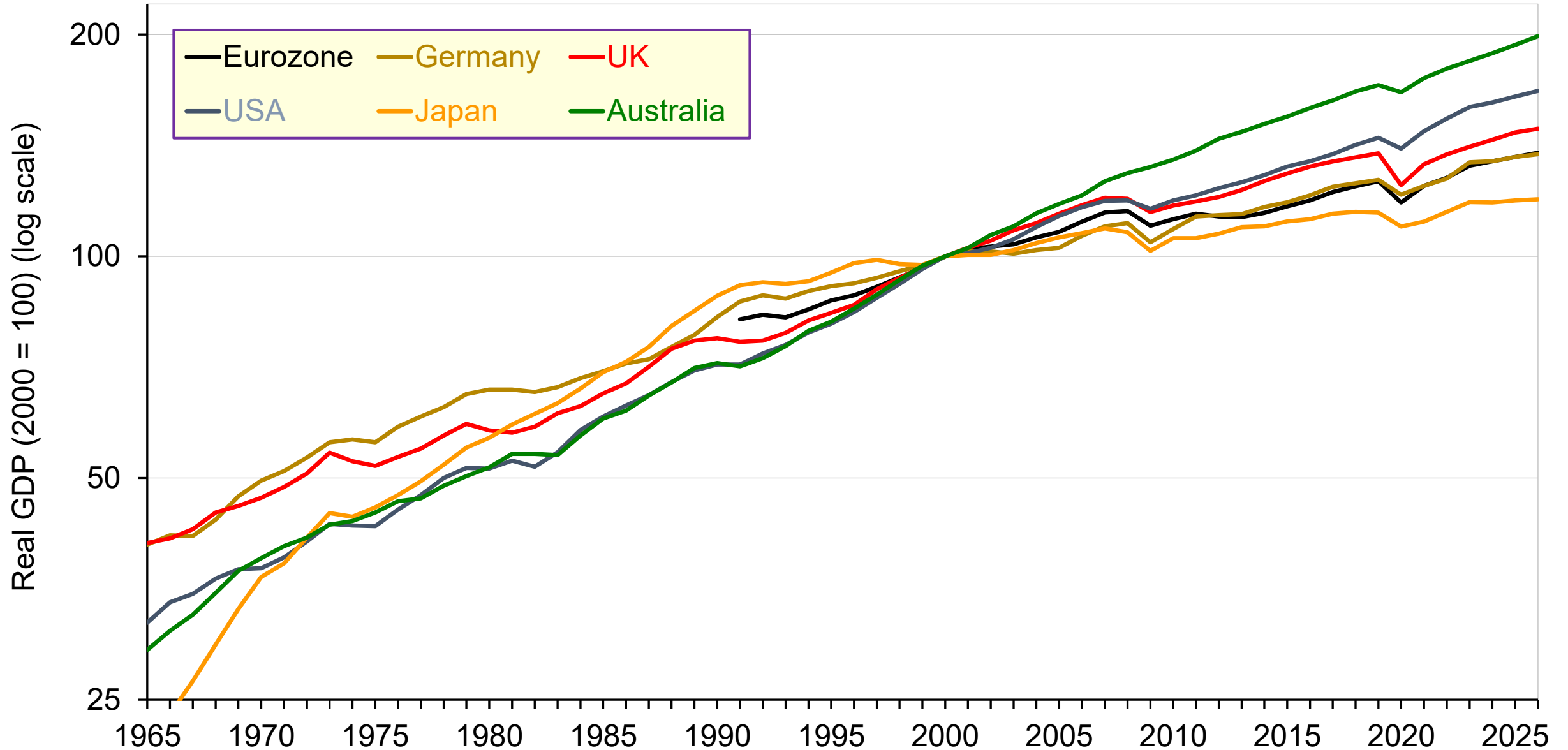
Long-Term Economic Growth



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- Despite short-term fluctuations and cycles, most economies show a positive growth trend over decades.
- When plotted over time, the level of Real GDP generally trends upward.
- While the rate of growth may go up and down (The Business Cycle), the total volume of output (The Path) typically increases in the long run.

Output paths from 1965



Notes: Data from 2022 based on forecasts; Eurozone = 19 countries using the euro (as of 2022); figures for Germany based on West Germany only up to 1991.

Sources: Based on data in [AMECO Database](#) (European Commission) up to 1980; and [World Economic Outlook](#) (IMF, October 2022)

Unemployment

Unemployment

- Those of working age who are without work but are available for work at current wage rates.
- Unemployment is tied to the business cycle; high economic activity reduces it, while recessions increase it.
- Factors like rapid industrial change, technological advancement, and globalization also drive job losses.
- Government Objectives: Reducing unemployment is a primary goal because:
 - It prevents the waste of human resources
 - It reduces the drain on government revenues (benefit payments).

Q If there are 3 million people unemployed and 24 million people employed, the rate of unemployment will be:

- A. 3%
- B. 8%
- C. 9%
- ☒ D. 11.1%
- E. 12.5%

Unemployment

$$\text{Unemployment Rate} = \frac{\text{Unemployed People}}{\text{Labor Force}} \times 100$$

Where:

- **Unemployed People** = 3 million
- **Employed People** = 24 million
- **Labor Force** = Unemployed + Employed = 3+24=27 million

$$\text{Unemployment Rate} = \frac{3}{27} \times 100 = 11.11\%$$

Measuring Unemployment

Standardised (ILO) Rate:

- The globally accepted measure used by the ILO and OECD.
- Defined as people of working age who are available to start and actively seeking work.
- Data is collected via national Labour Force Surveys.

Claimant Count:

- Measures only those receiving unemployment-related benefits (e.g., Universal Credit in the UK).
 - Pros: Easy and inexpensive to collect.
 - Cons: Highly sensitive to changes in government eligibility rules; excludes those looking for work but ineligible for benefits.

The ILO measure is preferred for international and historical comparisons.

Cyclical Patterns

- The Mirror Effect: Unemployment movements mirror the business cycle.
- Historical Trends:
 - Rising Unemployment: Seen during recessions (early 80s, early 90s, 2008–09).
 - Falling Unemployment: Seen during boom years (late 80s, late 90s, mid-2000s).

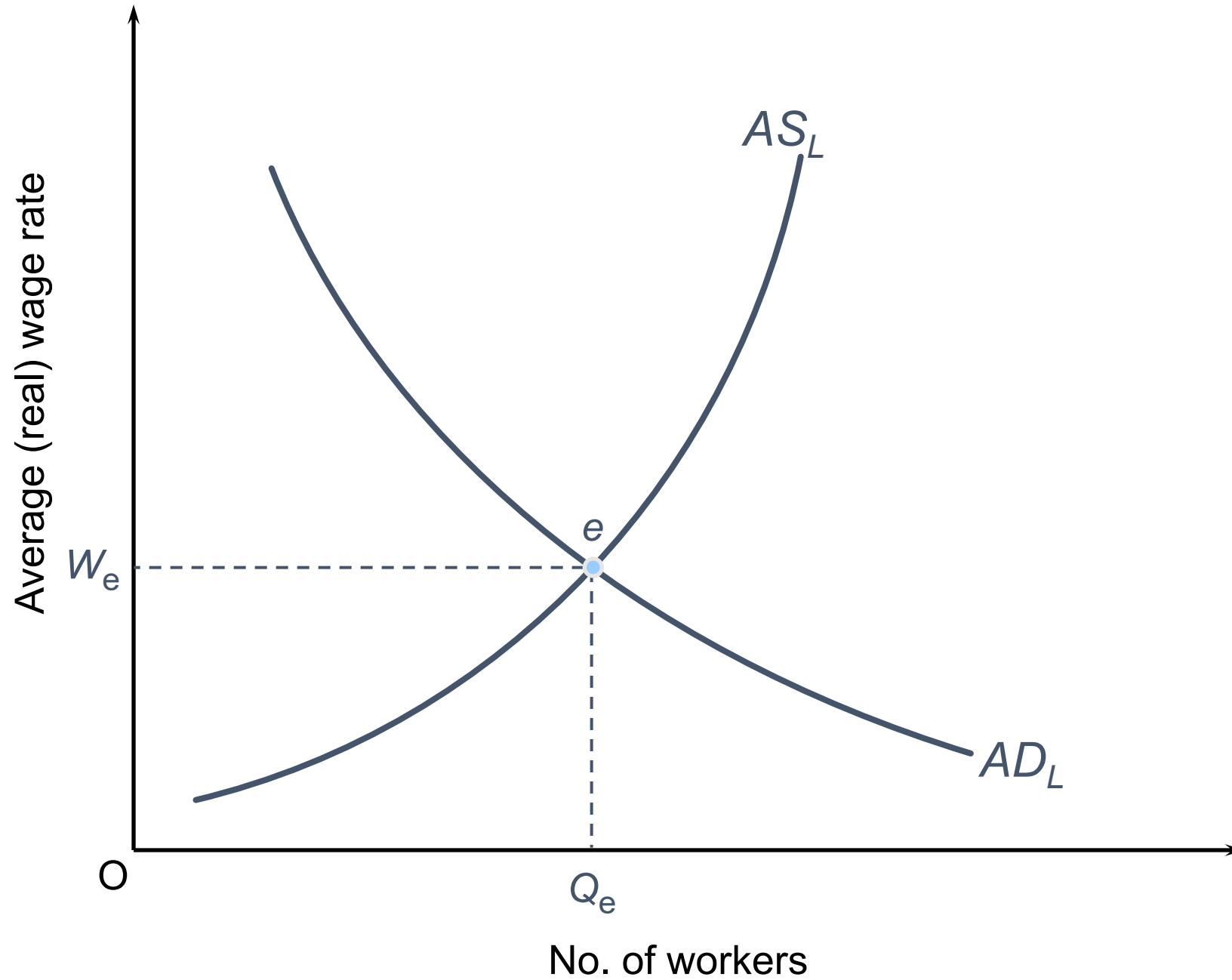
The Costs of Unemployment

- Individual Costs: Loss of earnings, lower self-esteem, and stress-related illnesses.
- Social Costs: Strained family relations and potential increases in domestic instability.
- Economic Costs:
 - Lost Output: Under-utilization of resources leads to lower national income.
 - Tax Loss: Reduced revenue from income tax, national insurance, and VAT.
 - Public Spending: Increased costs for healthcare, police, and social services.
- A Silver Lining? Short-term voluntary unemployment can lead to better job matching and a more mobile, efficient workforce.

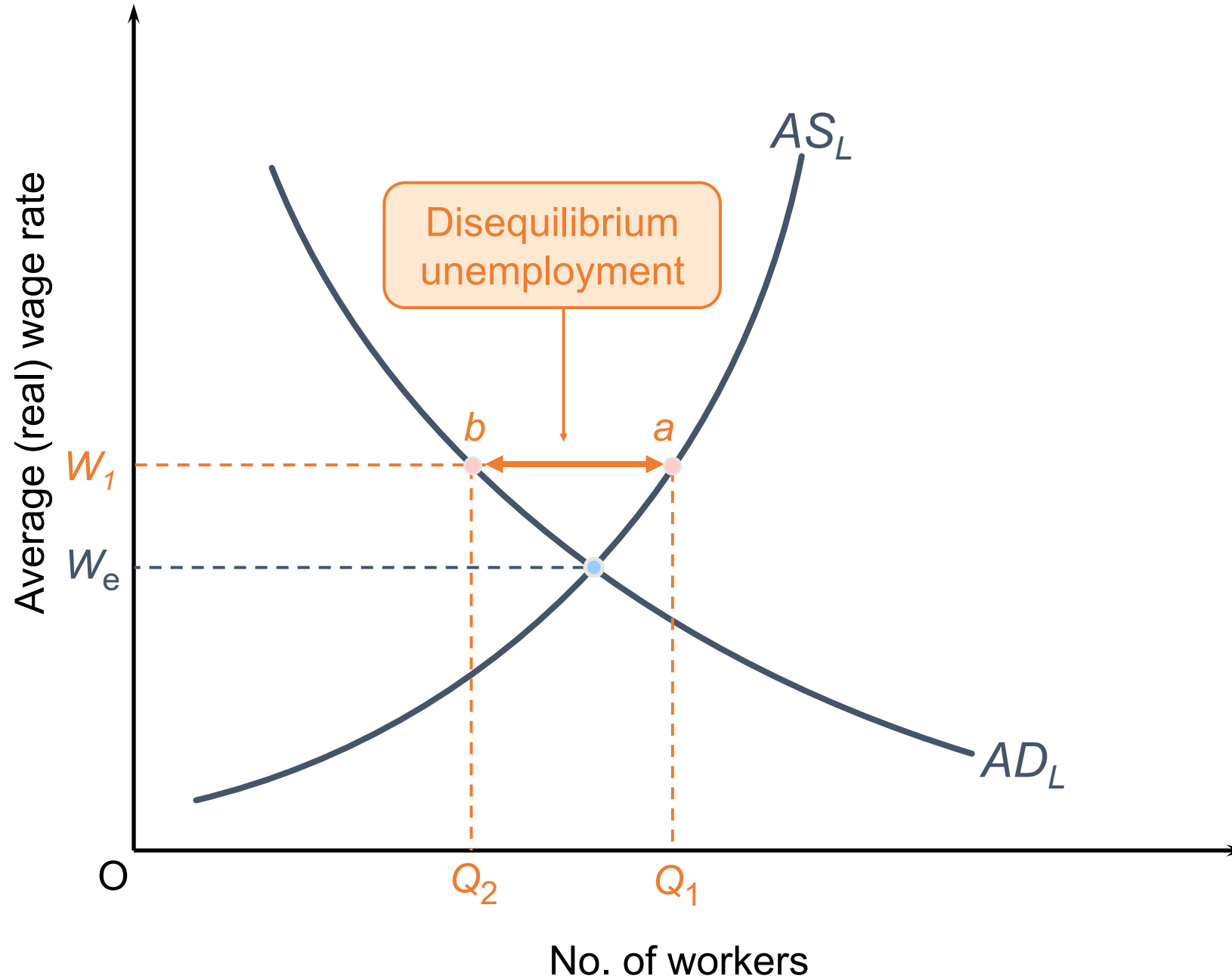
The Labour Market Model

- The Two Categories: Unemployment is classified as either Equilibrium or Disequilibrium.
- Market Dynamics:
 - ASL (Aggregate Supply of Labour): Number of workers willing to work at a given real wage. Relatively inelastic.
 - ADL (Aggregate Demand for Labour): Number of workers firms want to hire; slopes downward as higher wages encourage firms to substitute labour for machinery.
- Disequilibrium: Occurs when the real wage (W_1) is stuck above the equilibrium rate (W_e), creating an excess supply of labour.

Equilibrium unemployment



Disequilibrium unemployment



Types of Disequilibrium Unemployment

- Real-Wage Unemployment: Caused by wages being driven above market-clearing levels by trade unions or national minimum wages.
- Demand-Deficient (Cyclical) Unemployment: Occurs during recessions when consumer demand falls.
 - Firms cut production and reduce staff; the ADL curve shifts left.
- Growth in Labour Supply: If the workforce grows (e.g., school leavers) faster than jobs are created, and wages are sticky downwards.

Equilibrium Unemployment

- Frictional (Search) Unemployment: The gap between leaving one job and finding another.
 - Driven by imperfect information; employers and workers need time to find the right match.
- Structural Unemployment: Caused by fundamental changes in the economy:
 - Changes in Demand: Shifts in consumer tastes (e.g., moving from coal to green energy).
 - Technological Unemployment: New labour-saving techniques (e.g., ATMs replacing bank tellers).
 - Regional Unemployment: When dying industries are concentrated in specific areas (e.g., Welsh coal mining).
- Seasonal Unemployment: Fluctuations based on the time of year (e.g., tourism in Cornwall or agriculture).

Q Frictional unemployment is the result of:

- A. a shift in the pattern of consumer demand.
- B.

 workers and employers being ill-informed about the labour market.
- C. the introduction of new technology.
- D. the economy entering the recessionary phase of the business cycle.
- E. employers responding to the time of year and cutting back on their level of production.

Frictional unemployment

- **Frictional unemployment** occurs when workers are **temporarily unemployed** while transitioning between jobs, entering the workforce for the first time, or searching for a better job opportunity.
- A key reason for frictional unemployment is **imperfect information** in the labor market, meaning workers and employers are not immediately aware of all available job openings or suitable candidates.
- This **lack of information** leads to a delay in finding the right job match, causing short-term unemployment.

Frictional unemployment

- A. This describes **structural unemployment**, where industries decline due to changing demand, leading to job losses in certain sectors.
- C. This also relates to **structural unemployment**, as automation or new technology can replace jobs, requiring workers to retrain.
- D. This describes **cyclical unemployment**, which happens when economic downturns cause businesses to reduce their workforce.
- E. This describes **seasonal unemployment**, which occurs in industries like agriculture or tourism where demand fluctuates with seasons.

Option B is correct because frictional unemployment is caused by a lack of information, leading to delays in matching workers with job opportunities.

Inflation

Inflation

- A general and sustained rise in the level of prices across the entire economy.
- The Government's Goal: To keep inflation low and stable.
- Justification: Stability fosters confidence, allowing firms to set prices, negotiate wages, and make long-term investment decisions more effectively.
- The Link to the Business Cycle:
 - Rapid Growth: High demand often leads firms to raise prices (inflation rises).
 - Recession: Falling demand and rising stocks make firms unwilling to raise prices (inflation falls).

- The Role of Central Banks:
 - Institutions like the Bank of England and the ECB are tasked with meeting specific inflation targets.
 - UK Target: 2% for consumer prices.
 - Tool: Central banks adjust interest rates to keep inflation on track.

Measuring Price Changes

- Inflation vs. Deflation:
 - Inflation: Annual percentage increase in prices.
 - Deflation: A negative inflation rate where general prices are falling.
- Consumer Price Index (CPI):
 - The most common measure, focusing on the prices of goods and services bought by typical households.
 - Published monthly as a 12-month percentage change.
- The GDP Deflator:
 - A broader measure of inflation.
 - Covers the rate at which the prices of all domestically produced goods and services are changing.

Q Suppose that the CPI in a country increases from 150 to 153 over a period of a year. What is the annual rate of inflation?

A. 3%

☒ B. 2%

C. 1.02%

D. 0.03%

E. 0.02%

The annual rate of inflation:

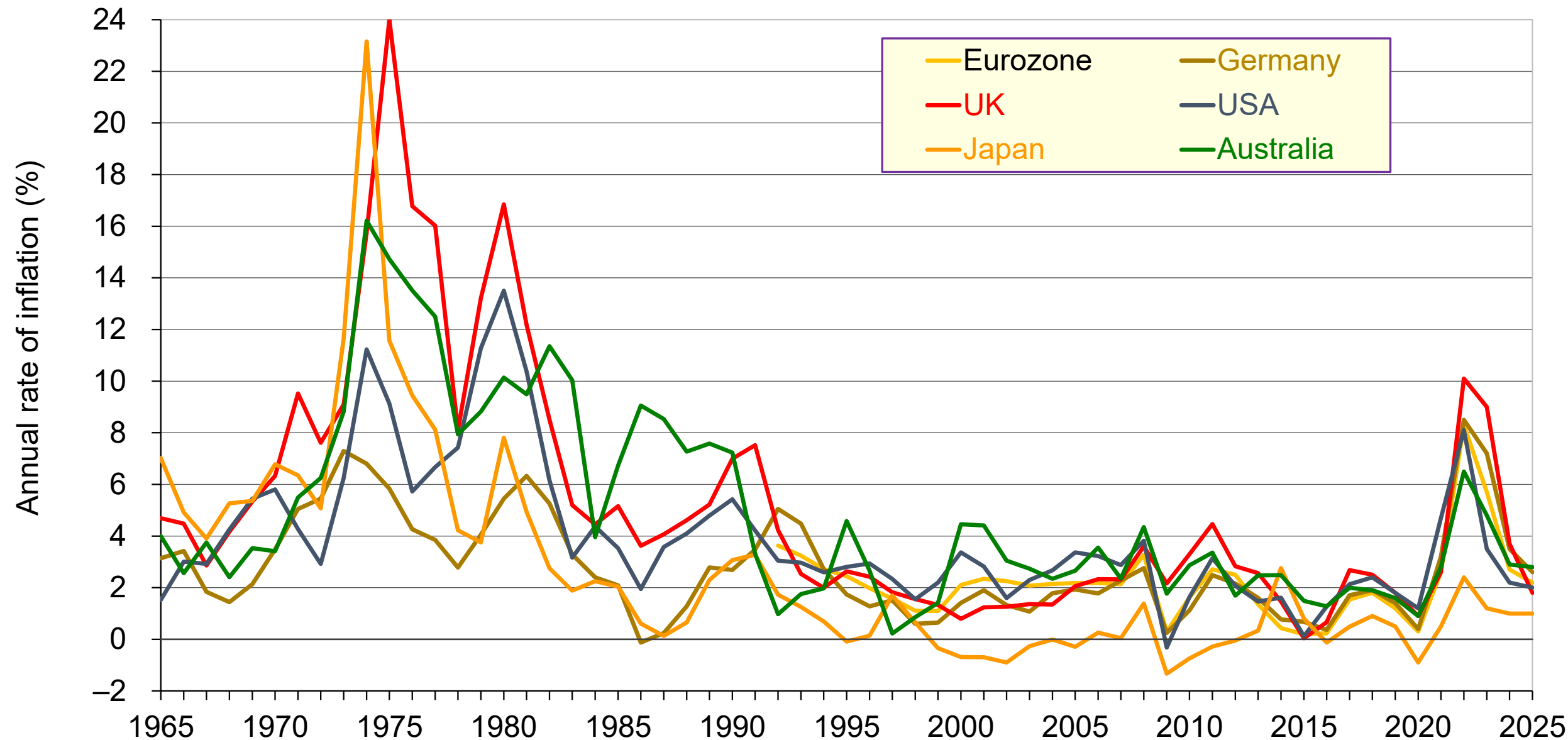
- The annual rate of inflation is calculated using the Consumer Price Index (CPI) formula:

$$\text{Inflation Rate} = \frac{\text{New CPI} - \text{Old CPI}}{\text{Old CPI}} \times 100$$

- **Old CPI = 150**
- **New CPI = 153**

$$\text{Inflation Rate} = \frac{153 - 150}{150} \times 100 = 2\%$$

Inflation rates



Notes: Inflation rate is the annual percentage change in consumer prices; Data from 2022 based on forecasts; Eurozone = 19 countries using the euro (as of 2022); figures for Germany based on West Germany only up to 1991

Sources: Based on data in AMECO Database (EC) up to 1980; World Economic Outlook (IMF, October 2022); various forecasts

The Costs of Inflation

- Redistribution of Wealth:
 - Losers: Those on fixed incomes (e.g., pensioners), those in weak bargaining positions, and savers whose interest rates are lower than inflation.
 - Winners: Those with assets (like property) that rise in value or those with the power to demand high pay/rent increases.
- Uncertainty and Investment:
 - High inflation tends to fluctuate more, making costs and revenues unpredictable.
 - This uncertainty discourages business investment, which can stifle long-term economic growth.

The Costs of Inflation

- Balance of Payments:
 - High domestic inflation makes exports less competitive abroad.
 - Imports become relatively cheaper than home-produced goods.
 - Trade deficits increase, exchange rates may fall, or interest rates may be forced upward.
- Resource Waste: Firms must divert resources (hiring accountants or financial experts) specifically to manage the complexities and uncertainties caused by rising prices.

Q In a period of rapid inflation which of the following would be the least desirable store of wealth?

A. Vintage wine

B. Property

C. Money

D. Land

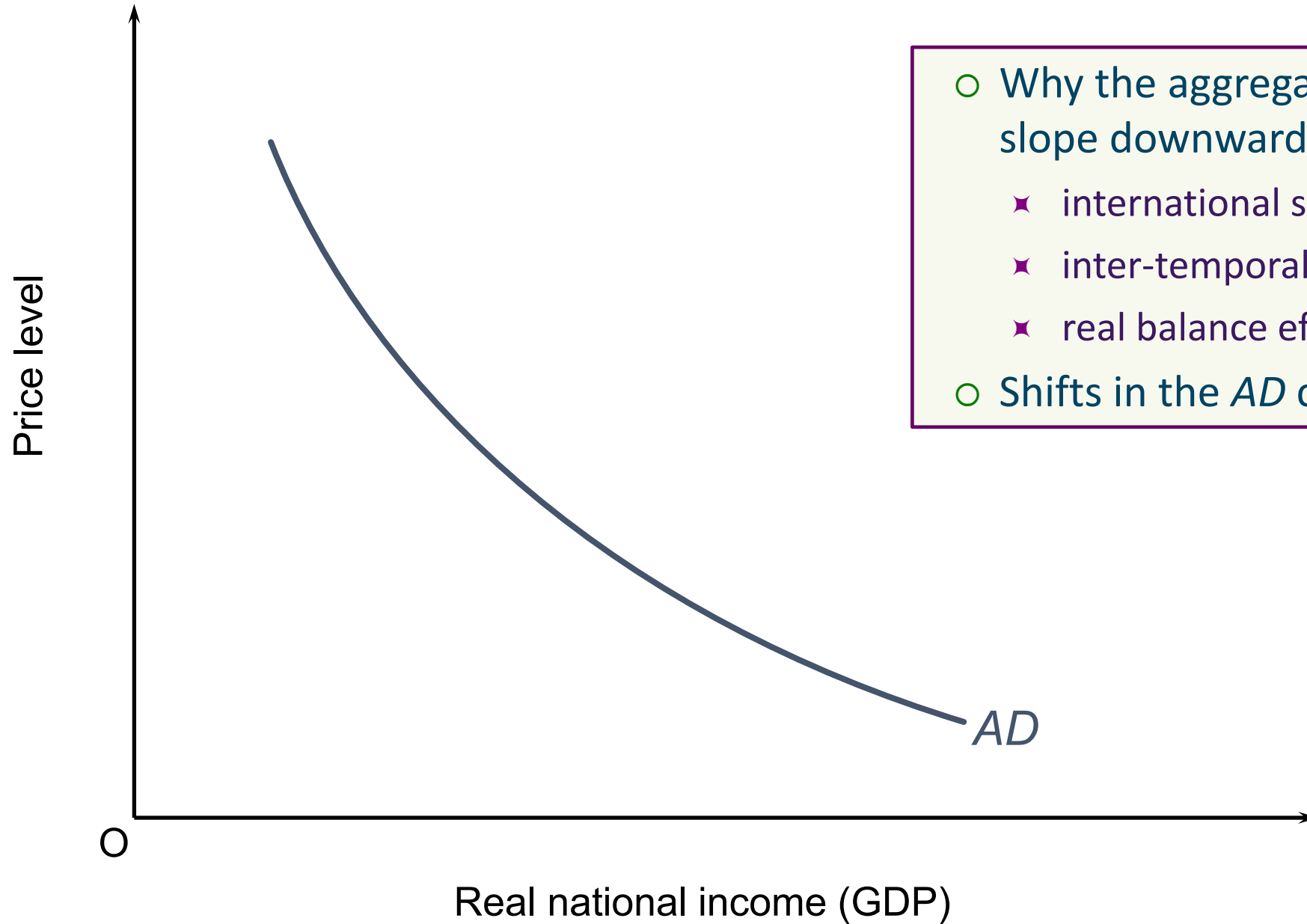
E. Stocks and shares

Money is the least desirable store of wealth during inflation because it loses value in real terms, while other assets can appreciate or maintain their worth.

Aggregate Demand (AD)

- The total national output (Real GDP) demanded at each price level (GDP Deflator).
- Why the AD Curve Slopes Downward:
 - International Substitution Effect: Higher domestic prices make exports expensive and imports cheaper, reducing demand for local goods.
 - Inter-temporal Substitution Effect: Higher prices increase the demand for money, driving up interest rates. This discourages borrowing and encourages saving.
 - Real Balance Effect: Inflation erodes the real value of savings, leading people to save more and spend less to compensate.
- Income Effect: If prices rise faster than nominal incomes, real income falls, forcing consumers to cut back on spending.

Aggregate demand

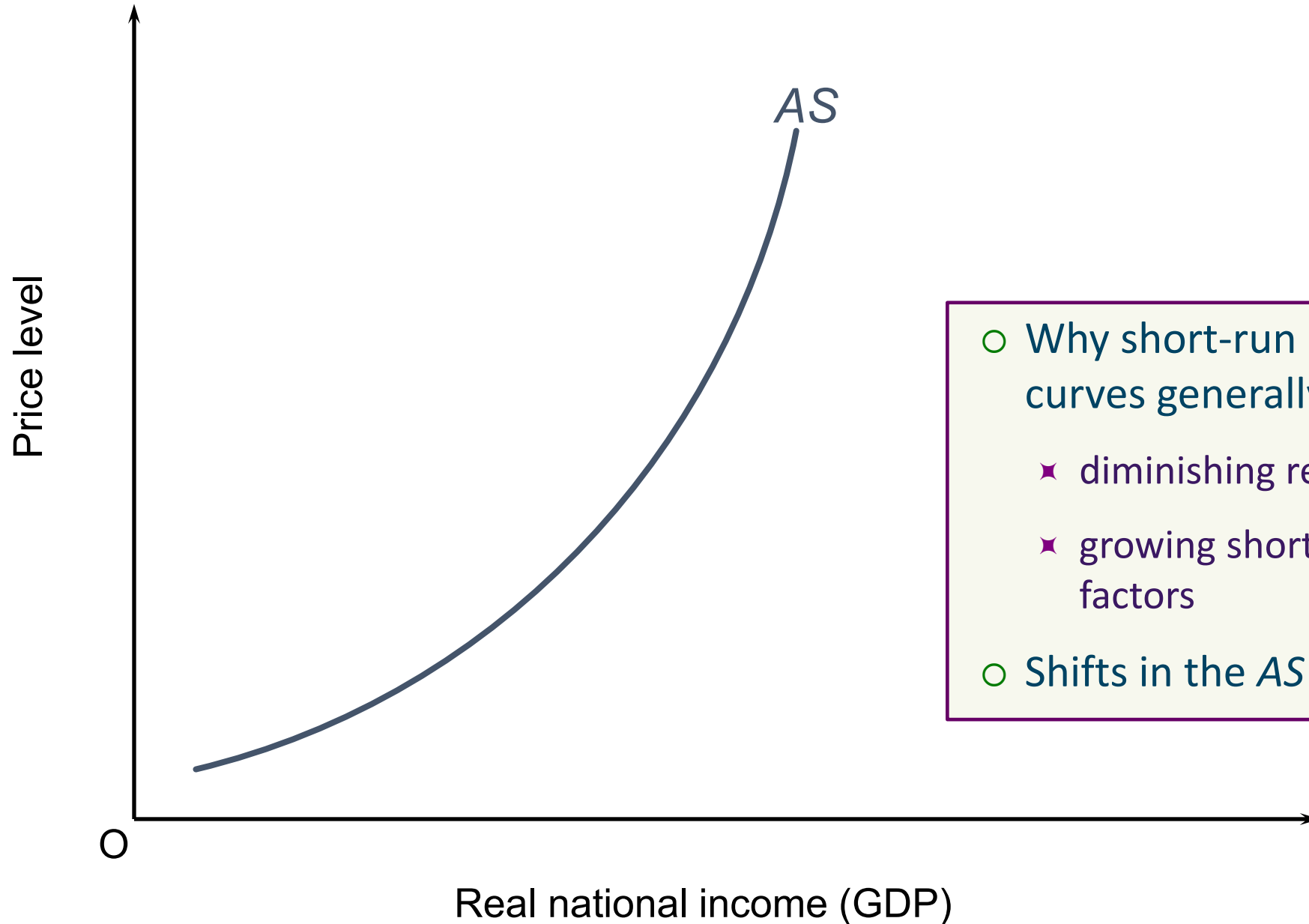


- Why the aggregate demand curves slope downwards
 - ✦ international substitution effect
 - ✦ inter-temporal substitution effect
 - ✦ real balance effect
- Shifts in the *AD* curve

Aggregate Supply (AS)

- Short-Run Aggregate Supply (SRAS): Shows the relationship between the price level and the quantity of output firms are willing to produce.
- The Upward Slope: In the short run, higher prices lead to higher production.
- Key Assumptions: Wage rates, input prices (raw materials), and technology are assumed to be constant.
- The Profit Incentive: Since input costs are fixed in the short run, rising product prices increase firm profitability, encouraging an increase in output.

Aggregate supply

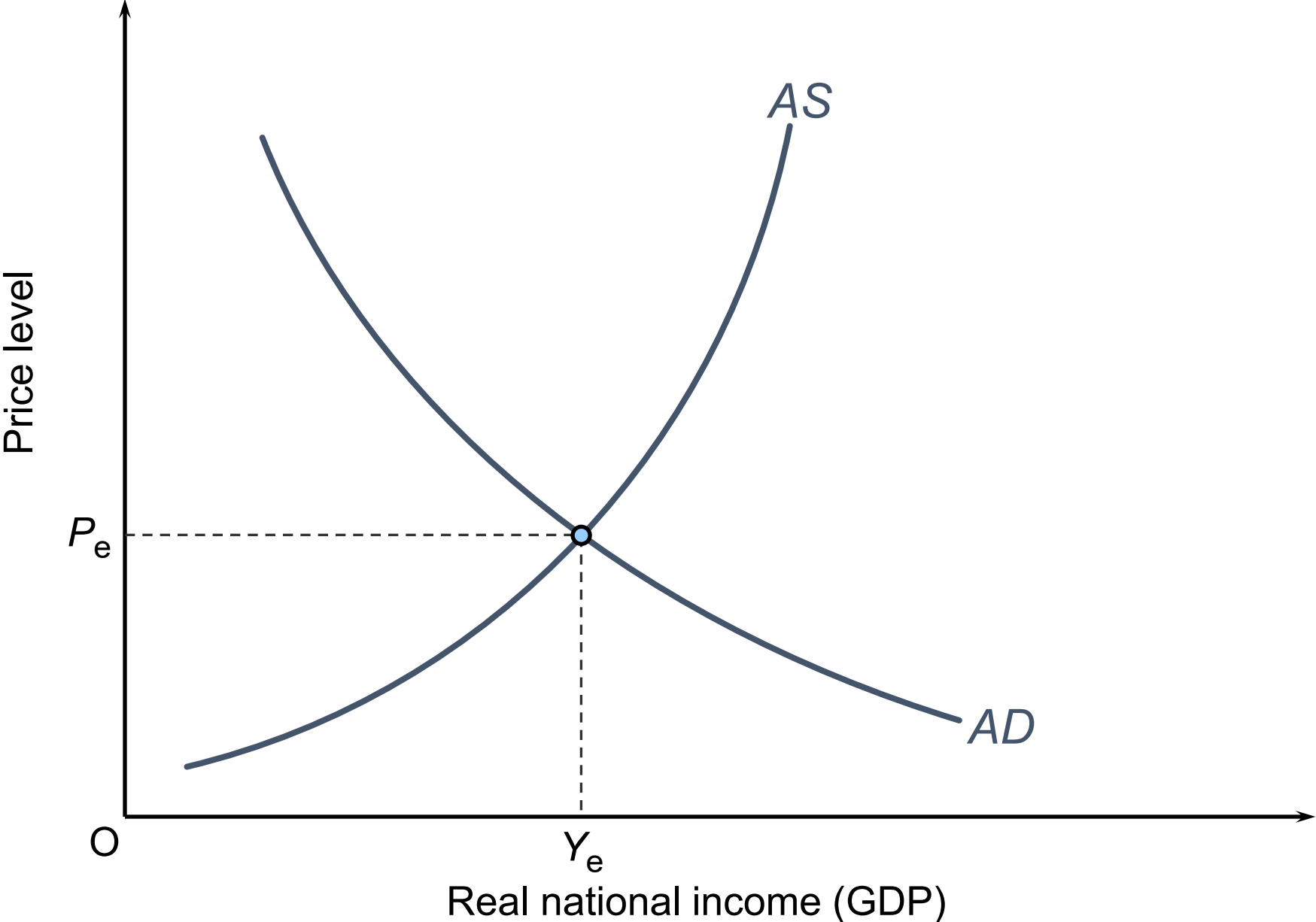


- Why short-run aggregate supply curves generally slope upwards
 - ✦ diminishing returns
 - ✦ growing shortage of certain variable factors
- Shifts in the AS curve

Macroeconomic Equilibrium

- The Equilibrium Point: Where Aggregate Demand equals Aggregate Supply ($AD = AS$).
- Market Correction:
 - If $AD > AS$, widespread shortages occur.
 - Shortages drive prices upward.
 - Rising prices cause a movement along both curves until they meet at the equilibrium price level (P_e).

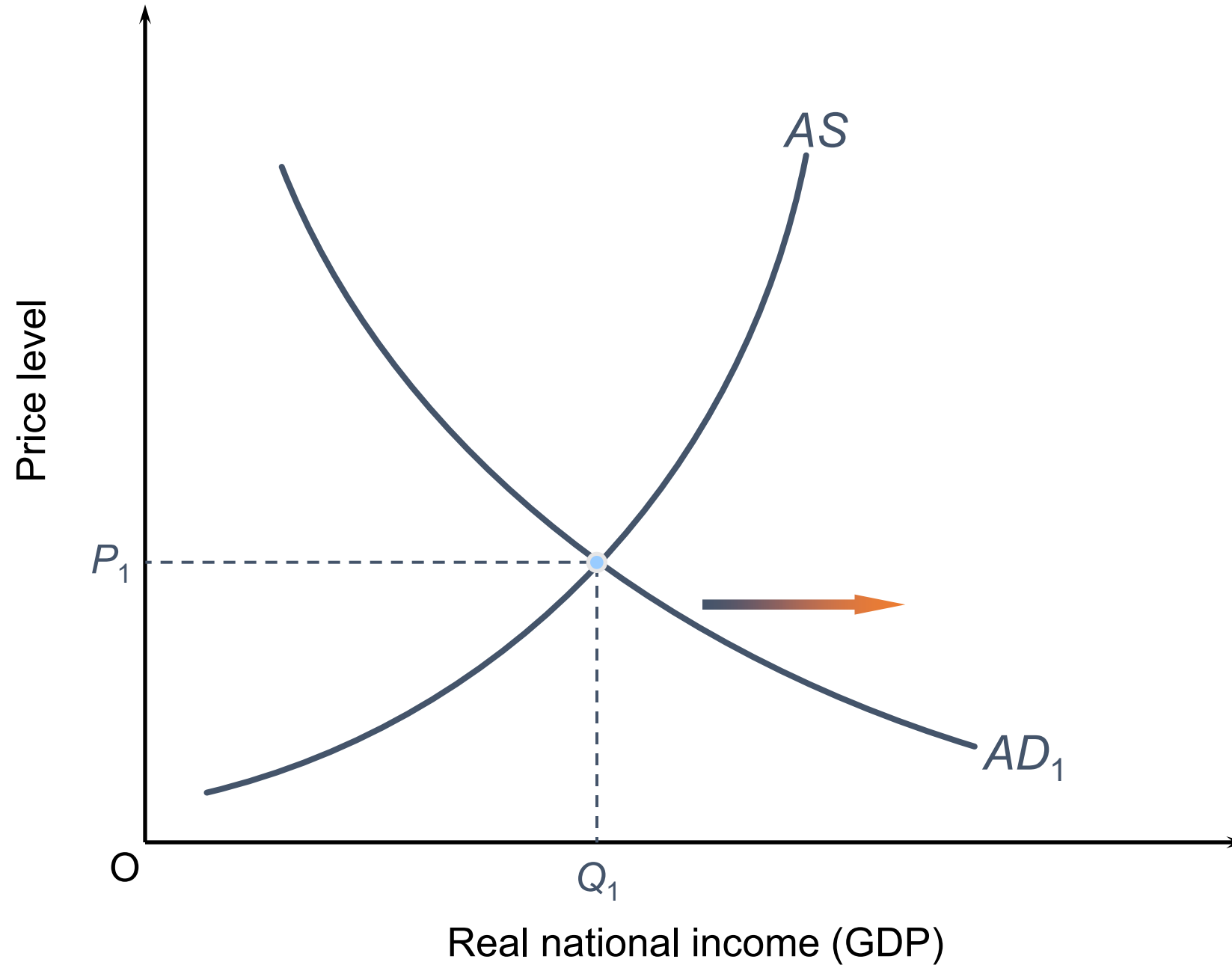
Aggregate demand and aggregate supply



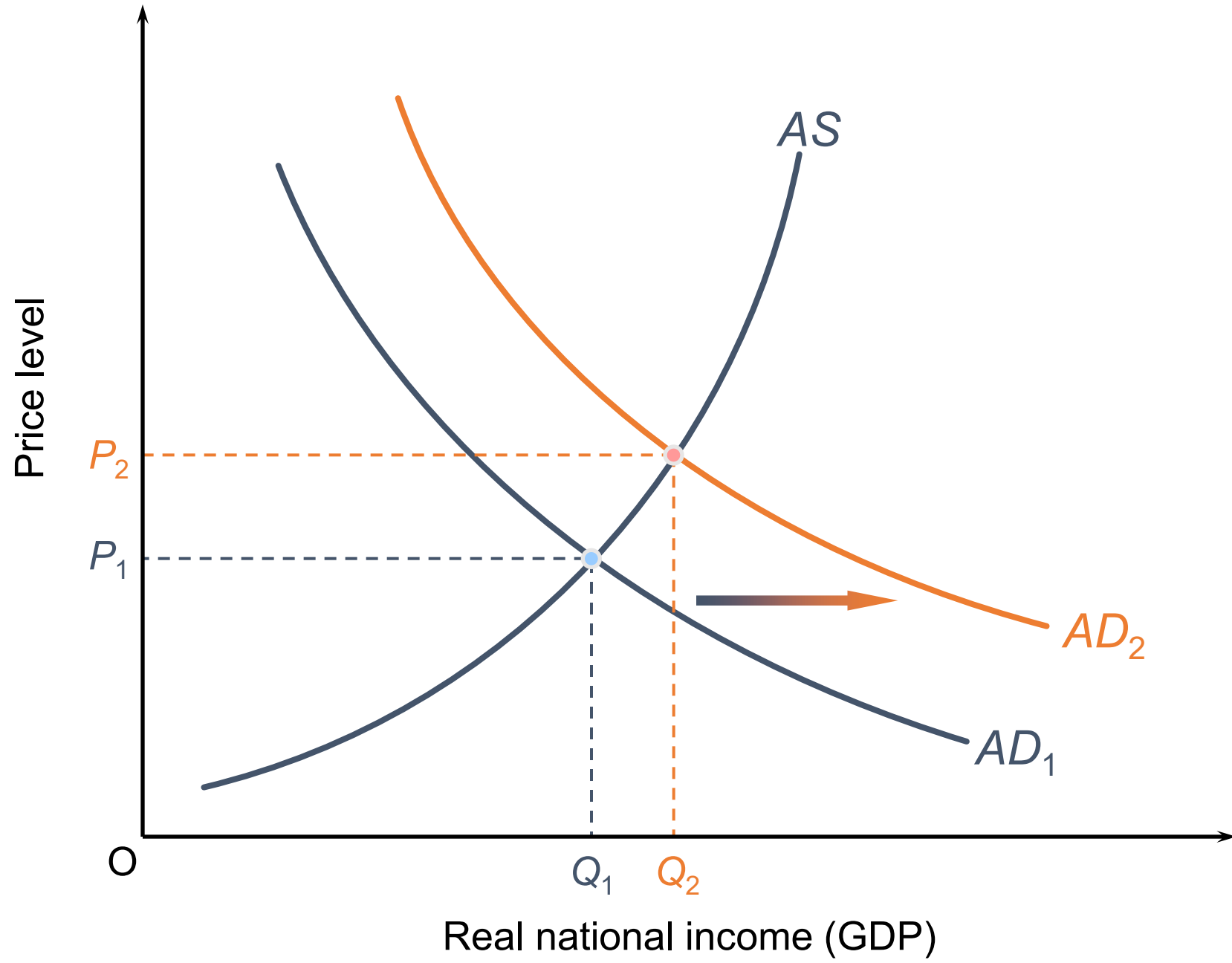
Causes of Inflation – Demand-Pull

- The Mechanism: Caused by persistent increases (rightward shifts) in Aggregate Demand.
- The Response: Firms react to high demand by increasing both prices and output.
- Capacity Constraints:
 - If there is slack in the economy (recession), prices rise slowly.
 - If the economy is near potential output (boom), prices rise sharply (steeper AS curve).
- The Trade-off: Demand-pull inflation is usually the counterpart to cyclical unemployment; when one is high, the other is typically low.

Demand-pull inflation



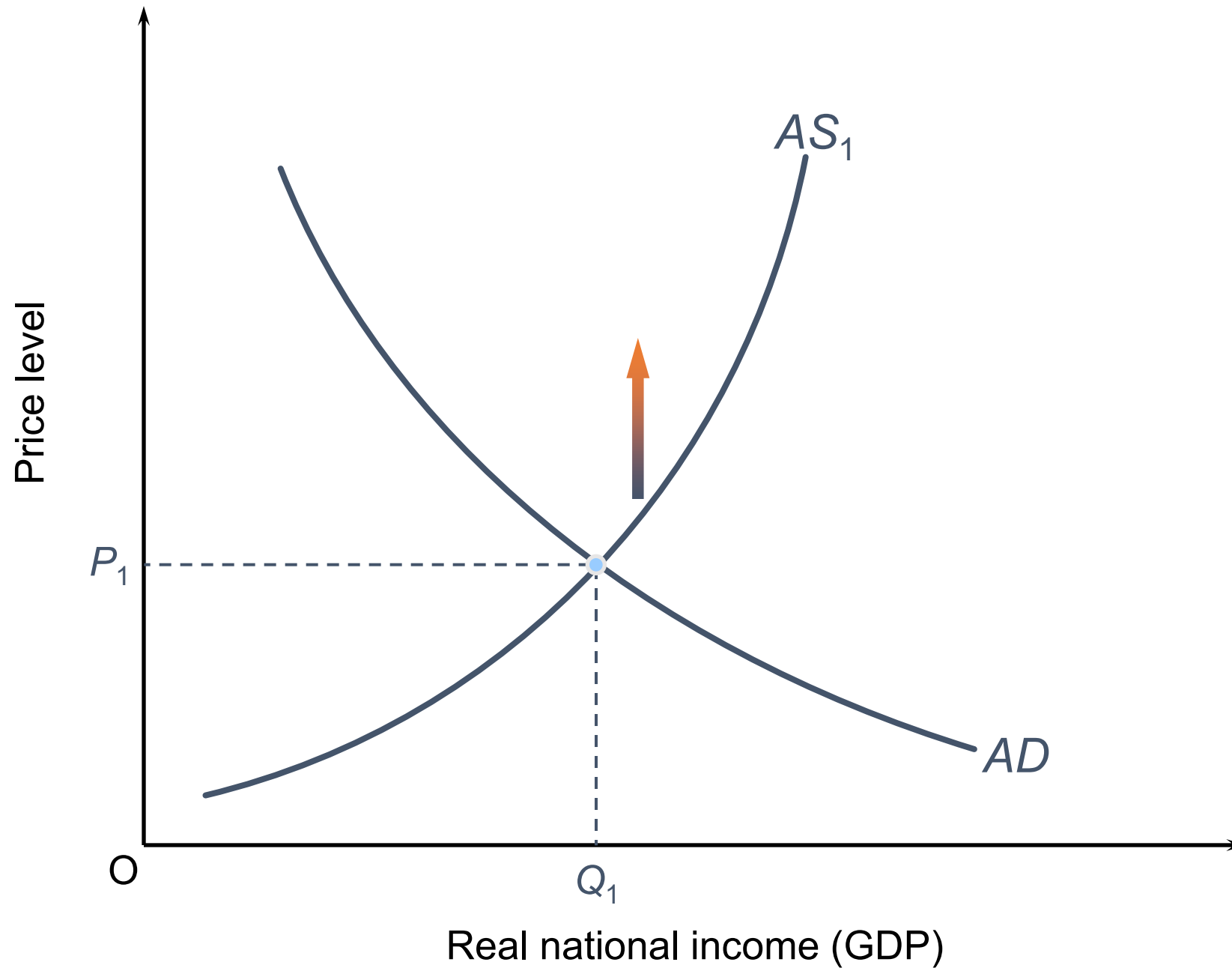
Demand-pull inflation



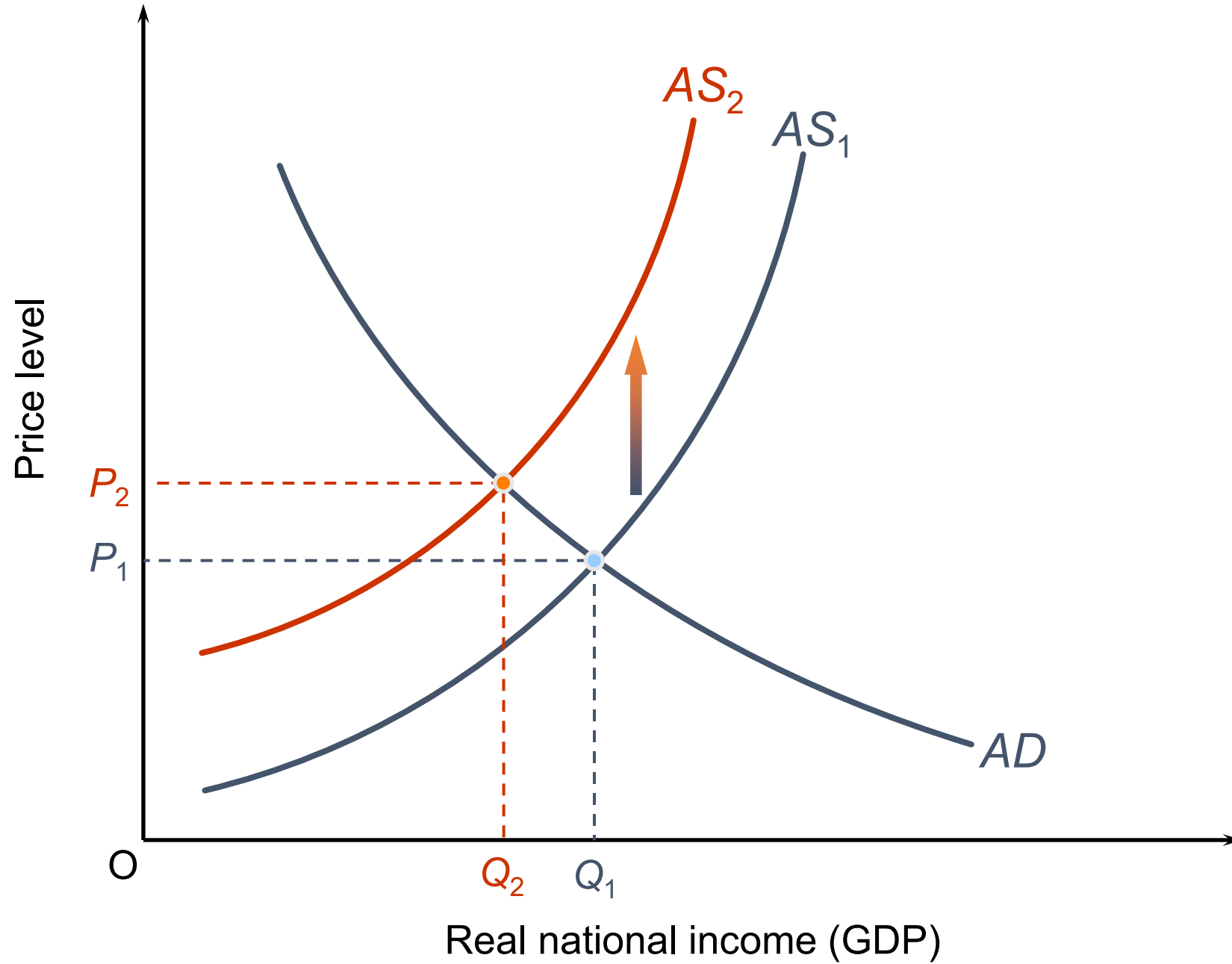
Causes of Inflation – Cost-Push

- The Mechanism: Caused by rising production costs independent of demand (e.g., rising oil prices or wages), shifting the AS curve to the left.
- The Impact: Firms pass on higher costs to consumers through higher prices while simultaneously cutting back on production.
- Role of Elasticity: If the AD curve is inelastic (consumers have few alternatives), firms can pass on more of the cost through higher prices with fewer lost sales.
- Demand-Pull vs. Cost-Push:
 - Demand-Pull: Output and employment tend to rise.
 - Cost-Push: Output and employment tend to fall (often leading to stagflation).

Cost-push inflation



Cost-push inflation



Q Which one of the following would be the cause of cost-push inflation?

A. A cut in the rate of income tax

B. A cut in the rate of VAT

C. A cut in interest rates

☒ D. A rise in the price of oil

Cost-push inflation

1. **A cut in the rate of income tax** – Increases disposable income, boosting consumer demand. This may cause **demand-pull inflation**, not cost-push inflation.
2. **A cut in the rate of VAT** – Lowers the price of goods and services, reducing inflation rather than increasing it.
3. **A cut in interest rates** – Encourages borrowing and spending, increasing demand, which leads to **demand-pull inflation**, not cost-push inflation.

A rise in oil prices is a **direct cost increase** for businesses, making it a **cost-push inflation factor**, whereas the other options influence demand rather than production costs.

Foreign Trade and Global Relationships

Foreign Trade and Global Relationships



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- **Global Interdependence:** A country's macroeconomic environment is shaped by both domestic conditions and its economic ties with the rest of the world.
- Shifting Global Order:
 - Rapid growth in emerging economies like China and India has fundamentally altered global trade patterns.
 - International cooperation (and shifts like Brexit) constantly redefine how nations trade and interact.
- **The Balance of Payments (BoP):** A record of all financial transactions between a country's residents and the rest of the world.
 - Debit Items: All payments made to other countries (outflows).
 - Credit Items: All receipts received from other countries (inflows).

Financial Well- being and Financialisation

Financial Well-being and Financialisation

- Financialisation: The increasing importance of financial markets, institutions, and products in the everyday lives of people and the health of nations.
- The Importance of Stability: The 2007–09 Financial Crisis proved that financial distress in one sector can spread globally like a contagion.
- The Debt Cycle:
 - Private sector spending weakened, forcing governments to spend more to offset the slump.
 - Falling tax revenues led to high government debt.
 - Resulting budget tightening (austerity) acted as a further brake on long-term economic growth.

Macroeconomic Policy Objectives

- **Growth:** Achieving high and stable economic growth.
- **Employment:** Maintaining low levels of unemployment.
- **Price Stability:** Keeping inflation rates low and predictable.
- **External Balance:** Avoiding large balance of payments deficits and volatile exchange rate fluctuations.
- **Financial Health:** Ensuring a stable financial system and preventing financial distress in households, firms, or the government.

Macroeconomic Policy Objectives



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- The Challenge of Policy Conflicts:
 - Objectives often pull in opposite directions, creating economic trade-offs.
 - Example: A policy to rapidly accelerate growth might inadvertently trigger higher inflation or lead to excessive borrowing and trade deficits.
- Success requires balancing these competing interests to create a sustainable macroeconomic environment.

Volatility and the Business Cycle

Economic Volatility and the Business Cycle

- A core feature of modern economies; they do not grow at a constant rate but experience expansions and contractions.
- Defining Recession: Officially defined as two or more consecutive quarters of declining output (negative growth).
- Actual vs. Potential Growth:
 - Actual Growth: The real-time percentage change in national output (GDP) over a year or quarter.
 - Potential Growth: The rate at which the economy could grow if it were using its resources at normal capacity.

Economic Volatility and the Business Cycle

Defining Capacity:

- Full Capacity: The absolute maximum production possible (working flat-out).
- Normal Capacity (Potential Output): A sustainable level that allows for planned spare capacity and natural job transitions.

The Output Gap: The difference between actual and potential output.

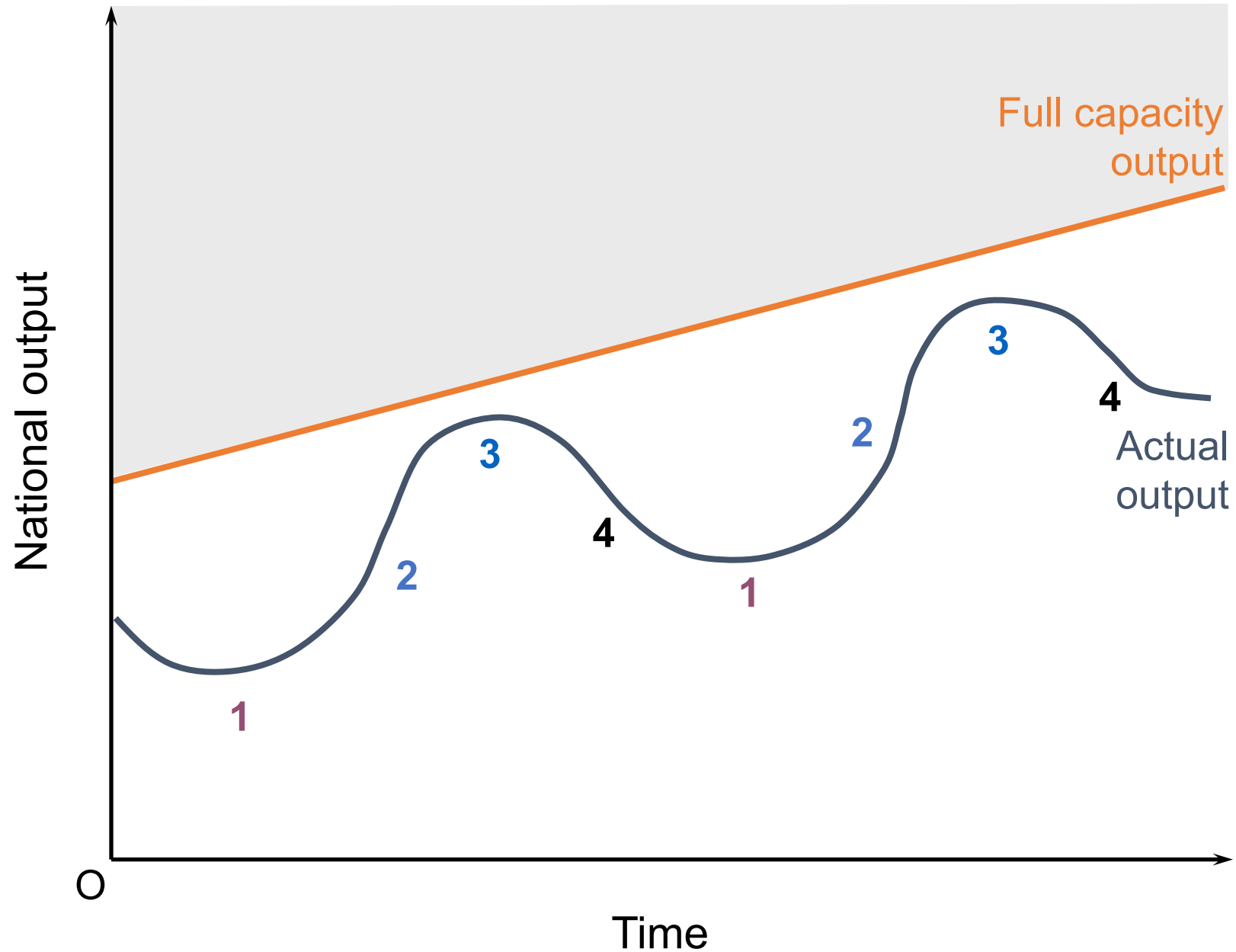
- Positive Gap: Actual output $>$ Potential (over-utilization, risk of inflation).
- Negative Gap: Actual output $<$ Potential (under-utilization, spare capacity, higher unemployment).

The 4 Phases of the Business Cycle

While every cycle is different, economists use a stylized model to identify four distinct phases:

1. The Upturn: A stagnant economy begins to recover; growth in actual output resumes.
2. Rapid Expansion (The Boom): Fast economic growth. The gap between actual and full-capacity output narrows as resources are used more fully.
3. Peaking Out: The rate of growth slows down or stops as the economy hits its ceiling.
4. Slowdown/Recession (The Slump): Little to no growth, or a contraction in output. Slack (unused resources) increases in the economy.

A hypothetical business cycle



The Business Cycle in Practice

- Irregularity vs. Theory: Real-world cycles are rarely as smooth or regular as the stylized models suggest.
- Variations in Length:
 - Booms and recessions can last from a few months to several years.
 - Double-dip Recession: When a recession is followed by a very brief recovery and then another immediate recession (e.g., the mid-1970s).
- Variations in Magnitude:
 - Growth Intensity: Phase 2 (Expansion) can range from explosive growth to a very gentle rise.
 - Severity of Phase 4: This phase might be a severe recession (like 2008–09) where output actually shrinks, or it may simply be a growth pause where growth remains positive but very low.

Aggregate Demand (AD) and its Components

- The total spending on domestically produced goods and services within a country.

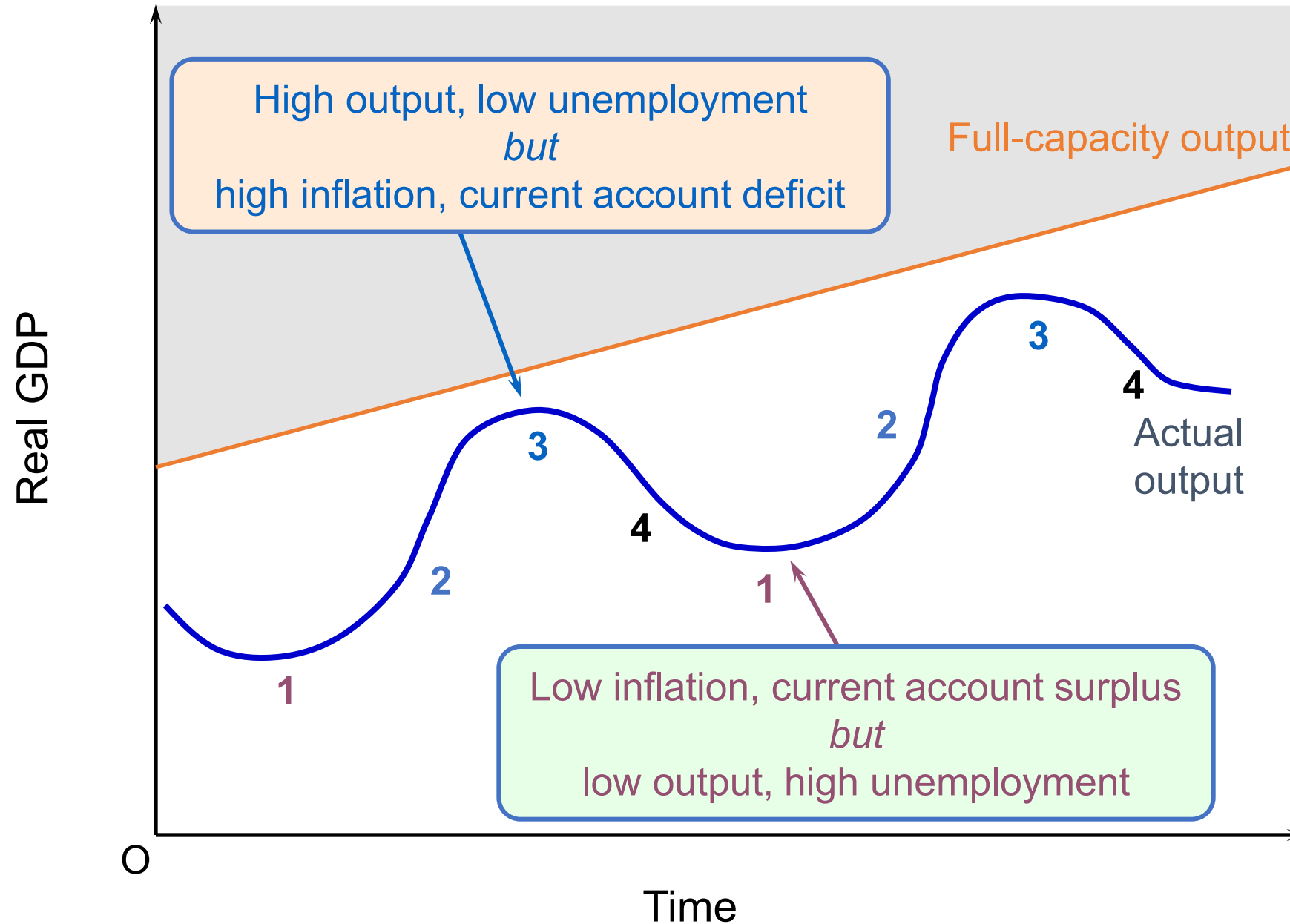
$$AD = C + I + G + X - M$$

- C: Consumer spending on goods and services.
- I: Investment spending by firms.
- G: Government spending on services and infrastructure.
- X: Exports (spending by people abroad on domestic goods).
- M: Imports (subtracted as this is spending on foreign production).
- Rapid AD growth creates shortages, stimulating firms to increase output (Boom).
- Falling AD leads to unsold stocks, causing firms to reduce output (Recession).
- In the short run, aggregate demand and actual output fluctuate together.

AD and Key Variables across the Cycle

- **Phase 2 (Rapid Expansion):**
 - **Unemployment:** Falls as firms hire more to meet rising demand.
 - **Inflation:** Rises due to shortages, especially near full capacity.
 - **Balance of Payments:** Deteriorates; high demand sucks in imports, and high prices make exports less competitive.
- **Phase 3 (The Peak):**
 - Output is at its highest and unemployment at its lowest.
 - Inflation and trade deficits are often at their most acute (severe) levels.
- **Phase 4 (Recession/Slump):**
 - Negative growth and higher **demand-deficient unemployment**.
 - Inflation usually slows down and the balance of payments tends to improve (though this may take time).
- **Financial Well-being:** During expansions, rising incomes allow agents to accumulate assets or reduce debt, though actual impact depends on individual behavior.

The business cycle and macroeconomic variables



Aggregate Supply (AS) and Expectations



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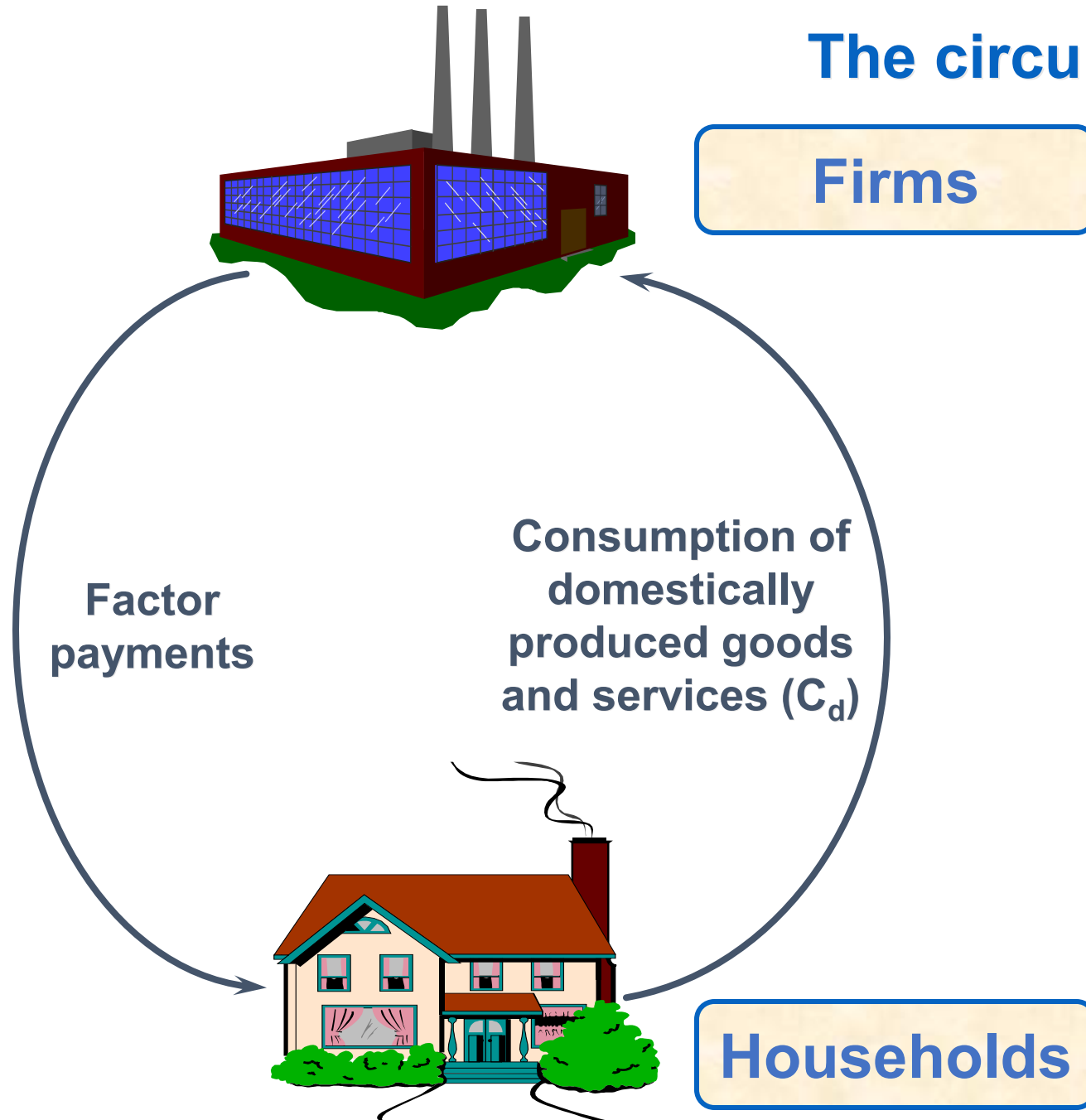
- The total amount of goods and services that firms plan to supply at current prices.
- Rising costs (e.g., higher wages or raw materials) can lead firms to reduce production
- Falling costs can encourage firms to increase production levels.
- The Power of Expectations:
 - If firms expect a downturn, they may scale back operations and cut investment in machinery or staff.
 - This can amplify a recession; a drop in AD is worsened by a cautious drop in AS.
 - Conversely, positive expectations can lead to higher current production and future investment.
- While AD drives the business cycle, AS is the primary driver of long-term potential growth.

The Circular Flow of Income

The Circular Flow of Income

- A simplified way to understand how economic choices affect short-term growth and Aggregate Demand.
- Two Main Groups:
 - Households: Owners of factors of production (labour, land, capital) and consumers of goods.
 - Firms: Producers of goods and employers of the factors of production.
- Stock vs. Flow:
 - Money (Stock): The total amount of currency in the economy at a specific moment.
 - Income (Flow): The amount of money moving between groups over a period of time.
- Velocity of Circulation: The speed at which money changes hands.
 - Example: If £1 trillion exists and moves twice a year, National Income = £2 trillion.

The circular flow of income



The Inner Flow and Withdrawals

The Inner Flow:

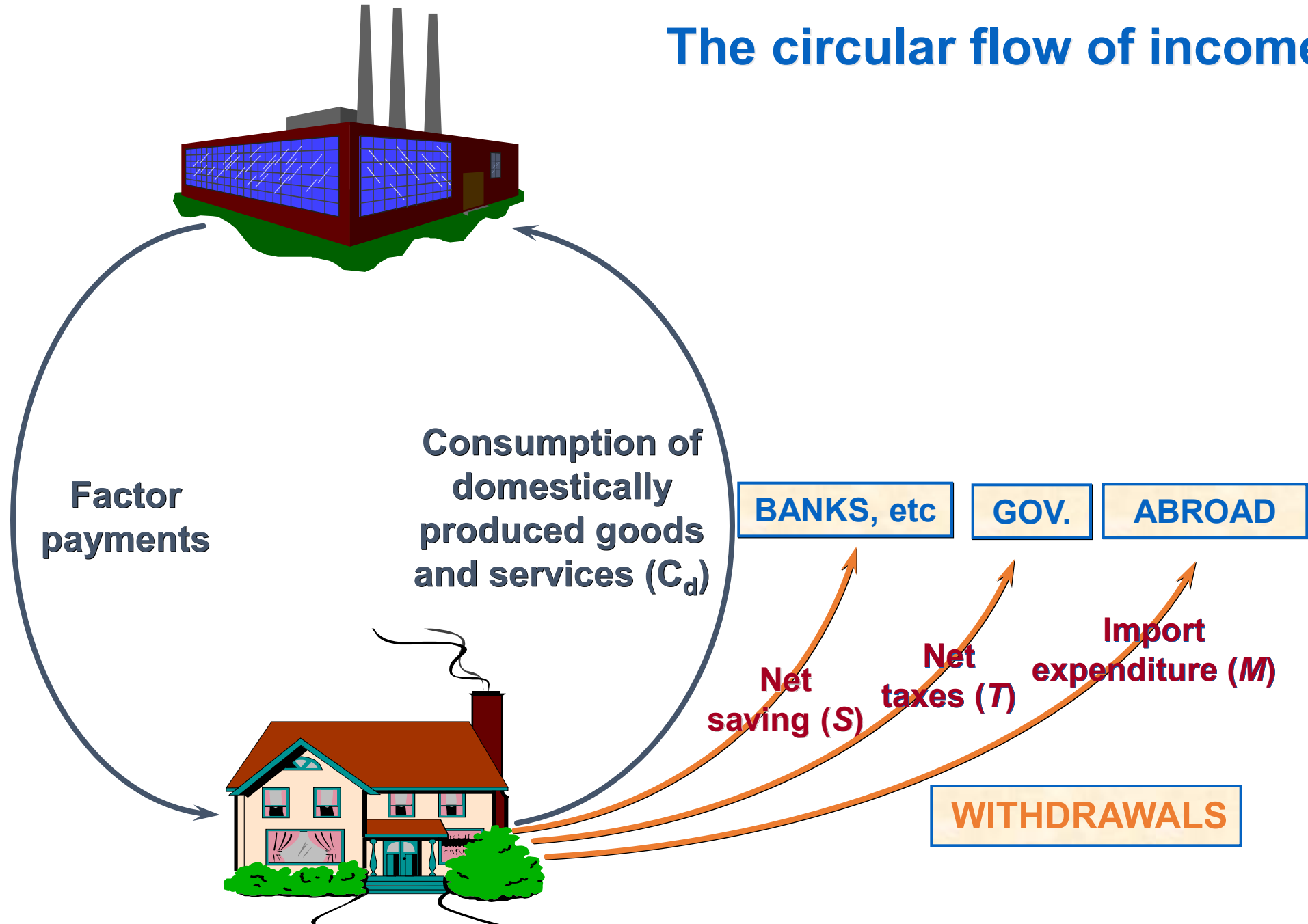
- Firms pay households (Factor Payments: wages, rent, dividends).
- Households pay firms for domestic goods (Consumption of domestic goods, C_d).

Withdrawals (W): Money that leaks out of the inner flow, reducing the circular movement.

1. Net Saving (S): Income put aside in banks minus any new household borrowing.
2. Net Taxes (T): Total taxes paid to the government minus Transfer Payments (benefits/pensions).
3. Import Expenditure (M): Spending that leaves the country to pay for foreign goods/services.

$$\text{Total Withdrawals Formula: } W = S + T + M$$

The circular flow of income



Injectons (J)

Money entering the circular flow from sources other than domestic household consumption.

- The Three Injections:

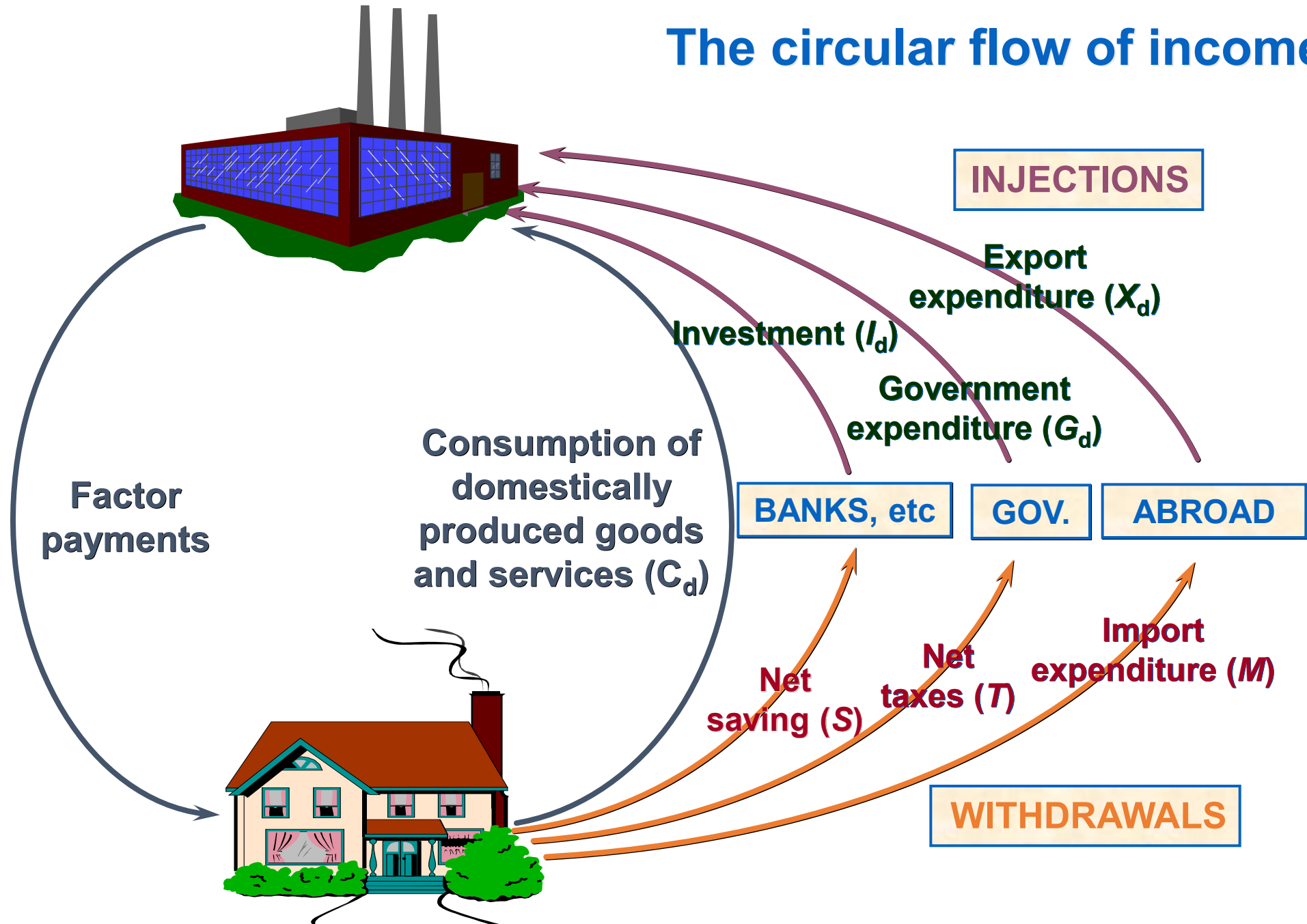
1. Investment (I_d): Spending by firms on capital (machines, buildings) and stocks, often funded by banks or shares.
2. Government Purchases (G_d): State spending on domestic goods and services (roads, schools, public sector salaries). Note: Excludes benefits.
3. Export Expenditure (X_d): Spending by foreign residents on domestically produced goods.

Total Injections Formula: $J = I_d + G_d + X_d$

$$AD = C_d + J$$

(Where C_d is consumption of domestic goods and J represents all injections).

The circular flow of income



Q Which of the following would represent a rise in aggregate demand?

- A. A rise in saving.
- B. A fall in investment.
- C. An increase in taxes.
- D. A fall in consumption.
- E.

 A rise in exports.

The Relationship Between W and J

While S relates to I, T to G, and M to X, they are governed by different sets of people and motivations:

- Saving vs. Investment: Households decide how much to save, while firms decide how much to invest. Their plans rarely match perfectly.
- Trade: Foreign demand for exports (X) does not automatically equal domestic demand for imports (M).
- Government: Governments may intentionally spend more than they collect in taxes (running a deficit) or vice versa.

The Reality of Disequilibrium: Because these decisions are independent, planned injections (J) frequently do not equal planned withdrawals (W).

Equilibrium in the Circular Flow

The state where Withdrawals = Injections ($W = J$).

The Adjustment Process (When $J > W$):

1. Increased Demand: If investment rises or taxes fall, Aggregate Demand increases.
2. Rising Output: Firms respond to higher sales by increasing production.
3. Rising National Income: To produce more, firms pay more in wages, rent, and profits to households.
4. Rising Withdrawals: As household incomes rise, people naturally save more (S), pay more tax (T), and buy more imports (M).

Equilibrium in the Circular Flow

5. New Equilibrium: Income continues to rise until withdrawals increase enough to exactly match the new level of injections.

The Reverse: If $W > J$, national income falls, leading to lower withdrawals until $W = J$ again.

Impact on Key Macroeconomic Variables

When Injections exceed Withdrawals ($J > W$), the resulting rise in Aggregate Demand has specific effects:

- Economic Growth: Actual national income rises; the larger the gap between J and W , the stronger the growth.
- Unemployment: Falls, as firms hire more labor to meet the surge in demand.
- Inflation: Tends to rise as the output gap closes and firms find it harder to meet demand without raising prices.
- Balance of Payments: Generally deteriorates; higher incomes suck in more imports, while inflation makes domestic exports less competitive.
- Financial Well-being: Rising national income allows households and firms to build up assets or pay down debts (improving balance sheets).

Economic Growth

Measuring and Comparing Growth

- Real GDP: Total national output.
- Real GDP per Worker: A measure of labor productivity.
- Real GDP per Capita: Total output divided by the population; the primary indicator of average living standards.
- The Impact of Small Differences: Even a 1% difference in annual growth rates leads to massive disparities in wealth when compounded over several decades.
- Population Factor: If population grows faster than GDP, living standards (GDP per capita) actually fall.

Average annual growth rates (%) 1961–2023

	Real GDP	Real GDP per worker	Real GDP per capita
Ireland	5.0	3.7	4.0
Japan	3.5	2.9	3.0
Portugal	3.1	2.9	2.9
Spain	3.2	2.4	2.4
Norway	3.0	2.0	2.3
Austria	2.6	2.1	2.2
Belgium	2.5	2.0	2.1
France	2.7	2.1	2.0
Italy	2.3	2.0	2.0
USA	2.7	1.4	2.0
Germany	3.0	1.6	2.0
Sweden	2.3	1.7	2.0
Netherlands	2.5	1.9	2.0
UK	2.3	1.7	1.9
Switzerland	2.2	1.1	1.3

Note: Figures from 2021 based on forecasts; German figures based on West Germany only up to 1991

Source: Based on data from [AMECO database](#) (European Commission, DGECFIN)

The Drivers of Labour Productivity

Since growth in output per worker is the engine of long-term prosperity, we must understand what drives Labour Productivity:

- Physical Capital (K):
 - The accumulation of machinery, equipment, and infrastructure through investment.
 - Better-equipped workers produce more in less time.
- Human Capital:
 - The collective knowledge, skills, and competencies of individuals.
 - Higher human capital leads to better idea generation and more efficient problem-solving.
- Technological Progress:
 - Innovation in digital tech, engineering, and materials.
 - Efficiency Gains: Modern technology allows a machine that costs the same as an old one to produce far more output, drastically increasing the productivity of both capital and labor.

Measuring National Income and Output

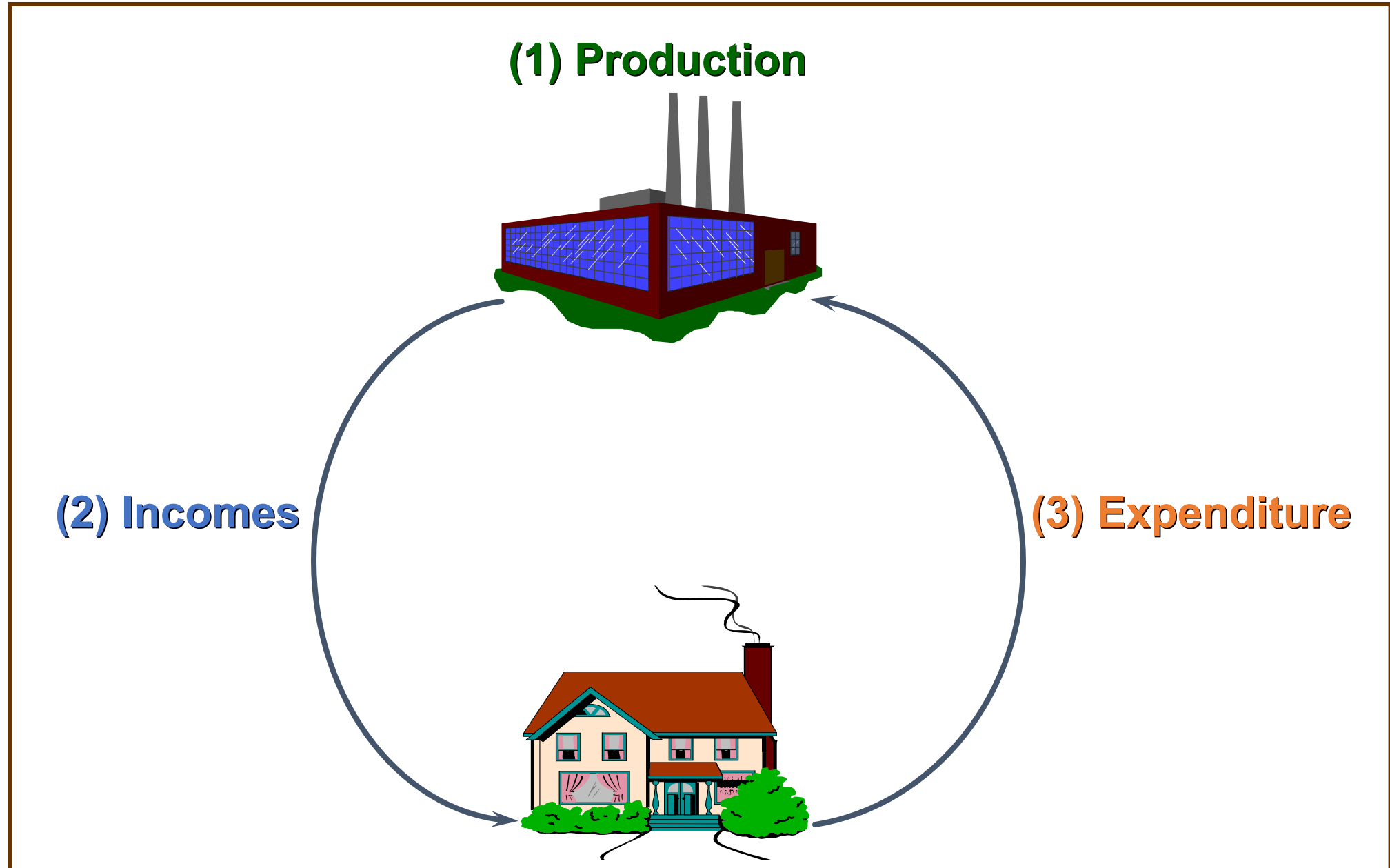
Measuring National Income and Output

- Gross Domestic Product (GDP) is the standard measure used to calculate the total value of a nation's output.
- GDP can be calculated using three different methods: Product, Income, and Expenditure.
- Because every pound spent on a product becomes someone's income and represents a portion of national output, all three methods should ideally result in the same total figure.

The Product Method (Value Added)

- Summing the value of all goods and services produced by every industry (Manufacturing, Construction, etc.).
- Avoiding Double Counting: We do not simply add up the total sales of every firm.
 - Example: If a manufacturer sells a TV for £200 and a retailer sells it for £300, GDP contribution is £300 (not £500).
- Gross Value Added (GVA): Calculated by adding the value added at each stage of production.
 - $\text{GVA at basic prices} = \text{Total Output} - \text{Intermediate Consumption (costs of purchases from other firms)}$.

The circular flow of national income and expenditure



UK GDP: 2020

UK GVA (product based measure): 2020	£m	% of GVA
Agriculture, forestry and fishing	12 578	0.6
Mining and quarrying;	13 347	0.7
Manufacturing	186 948	9.6
Electricity, gas, steam and air conditioning supply	30 396	1.6
Water supply, sewerage, waste management and remediation	24 123	1.2
Construction	112 624	5.8
Distribution, transport, hotels and restaurants	310 074	15.9
Information and communications	123 269	6.3
Financial and insurance	168 000	8.6
Real estate activities	266 621	13.7
Professional and support services	237 696	12.2
Total government, health and education	408 294	20.9
Other services and miscellaneous	55 635	2.9
GVA (gross value added at basic prices)	1 949 605	100.0

Source: *UK National Accounts, The Blue Book: 2021* (ONS, October 2021)

From GVA to GDP

- GVA is measured at basic prices, which excludes taxes and subsidies.
- GDP is typically reported at the prices consumers actually pay.

GDP at Market Prices = GVA + Taxes on Products (e.g., VAT) – Subsidies on Products

UK GDP: 2020

UK GDP: 2020	£m
GVA (gross value added at basic prices)	1 949 605

Source: UK National Accounts, The Blue Book: 2021 (ONS, October 2021)

The Income Method

- Summing all incomes generated from the production process.
- Components: Includes Wages and Salaries, Rent, Interest, and Profits (pre-tax).
- Why it Matches: Value added is effectively the difference between a firm's revenue and its external costs; this difference is distributed as income to those who provided the labor and capital.
- Exclusions: Transfer Payments (pensions, social security) are excluded because they do not represent new production of goods or services.

The Expenditure Method

- Adding up all spending on final domestic output.

$$\text{GDP} = C + G + I + X - M$$

- C (Consumption): Household spending on goods/services.
- G (Government): Spending on final goods and services (excludes transfers).
- I (Investment): Spending on capital assets and changes in inventories (stocks).
- X - M (Net Exports): Exports minus imports to ensure we only count domestic production.

UK GDP at market prices by category of expenditure: 2020

	£ million	% of GDP
Consumption expenditure of households and NPISH (C)	1 313 295	60.9
Government final consumption (G)	480 374	22.3
Gross capital formation (I)	360 958	16.7
Exports of goods and services (X)	600 973	27.9
less Imports of goods and services (M)	−596 693	−27.7
Statistical discrepancy	−2 834	−0.1
GDP at market prices	2 156 073	100.0

Q Given the following data about an economy (in £billions), what is the level of national income?

Saving 100, Consumption 350, Exports 200, Imports 250,
Net tax revenue 120 , Gov. expenditure 140, Investment 130

A. £470bn

B. £480bn

C. £570bn

D. £670bn

E. £820bn

GDP expenditure method

We use the expenditure method formula:

$$Y = C + I + G + (X - M)$$

Where:

- C = Consumption = 350
- I = Investment = 130
- G = Government Expenditure = 140
- X = Exports = 200
- M = Imports = 250

$$Y = 350 + 130 + 140 + (200 - 250)$$

$$Y = 570$$

We can also use the income method :

National income here is distributed in three main ways:

1. **Consumption (C)**: Spending by households.
2. **Savings (S)**: Income that is **not spent** but saved for future use.
3. **Taxes (T)**: Part of income that goes to the government.

$$Y=C+S+T$$

where:

- S = Savings (100)
- T = Net Tax Revenue (120)
- C = Consumption (350)

$$Y=350+100+120=570$$

Questions